How to Kernelize SVM

Example: Quadratic Kernel

$$\mathcal{L}_D(\alpha)=\sum_{i=1}^{N}\alpha_i-\frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\alpha_i\alpha_j y_i y_j X_i^T X_j$$

$$w=\sum_{i=1}^{N}\alpha_i y_i X_i$$

$$b=-\frac{1}{2}[\max_{i:y_i=-1} W^T X_i+\min_{i:y_i=+1} W^T X_i]$$

Class of vector $Z \Rightarrow \text{sign}(W^T Z+b)$

Kernelize using

$$K(a, b)=[a^T b]^2$$

$$\mathcal{L}_D(\alpha)=\sum_{i=1}^{N}\alpha_i-\frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\alpha_i\alpha_j y_i y_j [X_i^T X_j]^2$$

$$w=\sum_{i=1}^{N}\alpha_i y_i X_i$$

$$b=-\frac{1}{2}[\max_{i:y_i=-1} [W_i^T X_j]^2+\min_{i:y_i=+1} [W_i^T X_j]^2]$$

Class of vector Z is $\text{sign}([W^T Z]^2+b)$